

DEPARTMENT OF COMPUTER ENGINEERING
CSD415 MAIN PROJECT

Academic Year: 2026-27

Class: S7 C

PROJECT APPROVAL FORM

Date: 10 July 2026

Roll No	KTU Reg No	Name	Signature
2	CHN23CS004	Abel S Mathew	
5	CHN23CS013	Abraham Santhosh	
27	CHN23CS073	Diya Pauly	
44	CHN23CS128	Megha Ajith	

Name of Topic 1: Smart Contract Security Auditor

Abstract:

This project addresses the critical challenge of security vulnerabilities in blockchain smart contracts, focusing on the development of an autonomous, command-line interface (CLI)-driven developer tool to optimize vulnerability detection. The proposed system leverages an adversarial multi-agent Large Language Model (LLM) pipeline to scan Solidity source code for complex logic flaws such as reentrancy and access-control violations. By implementing a game-theoretic Multi-Agent Debate Protocol—where a Prosecutor Agent and a Defender Agent argue the validity of code defects before a Judge Agent issues a final security verdict—the solution drastically reduces false-positive noise compared to traditional static-analysis tools. Operating entirely within a high-performance terminal stream, the system streamlines developer workflows, provides automated code-remediation patches, and supports live on-chain contract scanning via remote blockchain APIs.

Ref: <https://doi.org/10.1109/TSE.2025.3597319>

Name of Topic 2: E-Commerce Scam & Fake Store Validator

Abstract:

This project addresses the critical challenge of retail cyber-fraud highlighted by law enforcement directives, focusing on the development of an automated, browser extension-driven validation tool to protect consumers from fraudulent Instagram-linked e-commerce storefronts. The proposed system leverages a four-module analytical backend to evaluate a target store's risk profile through domain-age/WHOIS analysis, semantic text-similarity scraping, localized product image pHash matching, and an authenticated UPI scam-ID trust registry. By integrating a weighted risk-scoring formula trained via a local logistic regression classifier, the solution synthesizes cross-module heuristics into a real-time, terminal-friendly advisory risk score without relying on live search engine scraping or external network overhead. Operating natively as a lightweight browser extension backed by a Python processing engine, the system streamlines digital safety workflows, eliminates data-access constraints, and provides a highly practical, software-only deployment framework optimized for developer usability.

Ref: <https://arxiv.org/pdf/2411.00368>

Name of Topic 3: Federated Learning for Blindness Detection

Abstract:

This project addresses data-privacy barriers and medical regulatory restrictions in healthcare tech, focusing on the development of a decentralized, privacy-preserving machine learning infrastructure for automated Diabetic Retinopathy screening. The proposed system leverages a classical Federated Learning (FL) server-client pipeline to collaboratively train deep convolutional neural networks (CNNs) across multiple simulated clinical nodes without centralizing raw medical images. By using the Flower orchestration framework, each independent client trains a lightweight neural network locally on isolated retinal fundus datasets, transmitting only mathematical parameter weight updates to a centralized server for global optimization using algorithms like FedAvg or FedProx. Operating as a pure software-only backend console application, this solution ensures strict data localization, prevents privacy leaks, and establishes a robust baseline for evaluating model convergence under heavily heterogeneous data distributions.

Ref: <https://arxiv.org/pdf/2411.00869>

Guide's Name:

Guide's Suggestion (if any):

Programme Assessment Committee (PAC):

1. Approved / Rejected
2. Approved / Rejected
3. Approved / Rejected

Selected Topic:

Project Co-Ordinator

HOD